

Defining DAX

30-Minute Hands-On Exercise | Week 9 - Section 4A

Learning Objectives: Define DAX and explain its role in Power BI | Describe how DAX differs from Excel expressions | Identify what DAX is uniquely suited for compared to Power Query | Distinguish DAX from Power Query across key characteristics | Apply the correct tool (DAX or Power Query) to real analyst scenarios

Instructions: Complete all four parts in 30 minutes. Refer to your Section 4A notes. Be ready to discuss your answers with the class.

Part 1: Fill in the Blank (~7 minutes)

Complete each sentence with the correct term or phrase from Section 4A.

1.	DAX stands for _____ .
2.	DAX is the native expression language of the in-memory columnar database engine called _____ , used wherever that engine is deployed.
3.	Unlike Excel, which uses cell coordinates to reference data, DAX uses _____ such as table names and column names to reference data in the model.
4.	DAX measures are calculated within a _____ , which is determined by the filters a user applies in a report -- making the result dynamic rather than stored.
5.	DAX is most useful for two types of analysis: _____ .
6.	Two other Microsoft products (besides Power BI Desktop) that use DAX are _____ .

Part 2: Matching (~6 minutes)

Write the letter of the correct description next to each term.

Term / Concept	Description
A. Expression Language	1. Summarizing many detail records into a single value such as a total, average, or count
B. Granularity	2. The set of filters active in a report at the moment a DAX calculation is evaluated
C. Filter Context	3. Power BI, Power Pivot for Excel, and SQL Server Analysis Services Tabular Mode

D. Aggregation	4. A dynamic DAX calculation that is recalculated based on filter context and is not stored per row
E. DAX Family of Products	5. A collection of functions, operators, and constants used in a formula to calculate or return values
F. Measure	6. The level of detail at which data is stored -- e.g., one row per transaction vs. one row per day

Part 3: DAX vs. Power Query Comparison (~8 minutes)

Place a checkmark (✓) in the correct column(s) for each feature. Some features apply to both tools.

Feature / Characteristic	Power Query	DAX
Can calculate column values		
Can calculate entire tables		
Can rename columns and tables		
Brings data into the model from external sources		
Structured as a series of sequential steps		
Uses nested expressions (not steps)		
Benefits from query folding for performance		
Can calculate dynamic measures within a filter context		
References table and column names (not cell coordinates)		
Best for dropping unnecessary data before loading		
Best for calculations that span data across multiple tables		
Can be reused across projects by cut-and-paste		

Hint: Review Sections 4A.6, 4A.7, 4A.8, and 4A.9. Some rows belong to both tools -- look carefully at the similarities listed in 4A.6.

Part 4: Short Answer (~9 minutes)

Answer each question in 2-4 sentences in your own words.

Question 1

A new analyst says: 'I already know Excel formulas -- DAX is basically the same thing, right?' How would you correct this misconception? What is the key difference in how DAX references data compared to Excel?

Question 2

A report needs to show each product's percentage share of total sales -- and that percentage must update automatically when a user filters by region or date. Why can this calculation NOT be built in Power Query? Which tool must be used, and why?

Question 3

Your Power BI model connects to a SQL Server database with 50 million rows. Only 2 million rows from the past year are needed for the report. Should you filter the data in Power Query or in DAX? Explain your reasoning using a concept from Section 4A.

Question 4

A colleague needs a calculation that looks at an order line item and determines what percentage of the total order that single line represents. Why is this type of calculation not suitable for storage in the dataset, and which tool is designed to handle it?

Bonus / Reflection (if time allows)

Section 4A states that 'how the data that DAX works with is shaped will drastically affect how useful our analytics can be.' In 3-5 sentences, explain what this means. What happens to DAX calculations when the underlying data model is poorly structured -- for example, when two fact tables are connected directly to each other instead of through a lookup table?